INCH-POUND
A-A-59939B
24 OCT 2014
SUPERSEDING
A-A-59939A
15 JUN 2014

COMMERCIAL ITEM DESCRIPTION

Testing Kit, Petroleum, Fuel Oxidation Tester

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes the requirements for a Jet Fuel Thermal Oxidation Tester (JFTOT) which adheres to the industry standard ASTM D3241 "Standard Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels (JFTOT Procedure)." The unit must meet the form, fit, and function of the salient characteristics listed below. The JFTOT is used to measure the high temperature stability of gas turbine fuels to conditions that can be related to those occurring in gas turbine engine fuel systems.

2. SALIENT CHARACTERISTICS.

2.1 Item Description. This is a Petroleum Test Kit Fuel Oxidation Tester. It a self-contained unit that tests and evaluates hydro-carbon based fuels to ASTM D3241 methodology and standards the characteristics of fuels produced according to the ASTM D1655 test methods in compliance with MIL-PRF-28800.

2.2 Instrument Classification. Petroleum Test Kit Fuel Oxidation Tester shall be class 3 test equipment and the enclosure shall be as provided by the manufacturer and conforms to the environmental requirements of class 3 in accordance to MIL-PRF-28800.

2.3 Sample Capacity. The fuel tester shall have a minimum sample capacity of 500mL (16.9 US ounces).

2.4 Fuel Oxidation Tester Analyzer Characteristics. The fuel tester shall be capable of testing fuels by supplying a minimum constant regulated mass flow ranging from 0 to 5 mL/min, (5 cm3/min, 5•10-3 L/min or 0.87987 quart/min or 0.21997 gallon/min).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC 404 SCMS/GUEEA, 235 Byron Street, Robins AFB, GA 31098-1670 or email at SPEC99@us.af.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/ .

AMSC N/A FSC 6630

Distribution Statement A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:31Z
Check the source to verify that this is the current version before use.

2.5 Fuel Oxidation Tester Flow Accuracy. The fuel tester shall have a flow accuracy range ≤ 3% from maximum flow.

2.6 Specific Gravity Range. The Fuel Oxidation Tester shall test and process samples with a minimum specific gravity test range between 0.60 to 0.99 relative to water, which is equal to 1 as shown in Table I.

Table I. Specific gravity & boiling temperature reference.

Specific Gravity Values of Common Hydrocarbons used by US Military			
Description	Temperature °F(°C)	Boiling(°C)	Specific Gravity(Water =1)
Diesel Fuel Oil 2D/3D/4D/5D	60(15.6)	260 to 315	0.81 - 0.96
Fuel oils 1/2/3/5A/5B/6		205 to 290	0.82-0.95
Gasoline A		−1 to 110	0.74
Gasoline B			0.72
Gasoline C			0.68
Jet fuel (JP8)		150 to 205	0.82
Jet fuel (A/A-1)			0.82
Kerosene		205 to 260	0.78-0.82
Naphthalene		30 to 200	0.76

2.7 Fuel Oxidation Tester Heat Characteristics. The Fuel Oxidation Tester shall test and process samples by applying a minimum heat to vaporize samples with a minimum specific heat (ΔHvap) range value varying from 236 kJoule/kg°C (100 BTU/lb.) to 2093 kJoule/kg°C (900 BTU/lb.).

2.8 Operating Temperature of Specimen. The Fuel Oxidation Tester shall apply to the specimen tube, heat of vaporization levels that achieve a minimum oxidation temperature range from 100°C (212°F) to at least 325°C (617°F) on demand as required by the user starting from an average controlled temperature of 25°C (77°F).

2.9 Operating Pressure (Test). The Fuel Oxidation Tester shall apply to the specimen tube, a pressure range from 680 kPa to 3400 kPa (100 psi to 500 psi) on demand by the user departing from 101kPa (14.7 psi).

2.10 Differential Pressure. The Fuel Oxidation Tester shall be capable of compensating for atmospheric variations up to 280mm Hg (5.4 psi).

2.11 Elapsed Time Test. The Fuel Oxidation Tester shall test and process samples within 1.15 hours ±0.25 hr. (average) while applying minimum heat to vaporize samples.

2.12 Analysis Thickness. The Fuel Oxidation Tester shall be capable of capturing precipitated deposits (residual samples) with thickness ranging from 10nm up to 1000nm on an area of 2mm2 to 3 mm2.

2.13 Deposits Spectra. The Fuel Oxidation Tester shall be capable of displaying the spectra of the non-volatile residues on heated tubes or surfaces for further spectral, chromatographic and tribology analysis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:31Z
Check the source to verify that this is the current version before use.

2.14 Design and Construction. The Fuel Oxidation Tester shall be used in a laboratory environment on bench top and mobile facilities. The Fuel Oxidation Tester is a commercial item subject to commercial products regulatory standards. The Fuel Oxidation Tester shall comply with all applicable MIL-PRF-28800F sections.

2.15 Commercial Conformance. The Fuel Oxidation Tester shall be designed and constructed in accordance with all applicable International & Industrial requirements to include the European Union (EU) requirements in order to have the "CE" marking affixed and awarded a certificate of conformance.

2.16 System Safety. The system shall comply with all sections of the Underwriters Laboratories (UL) standard 61010A and 61010B for electrical safety.

2.17 Electrostatic Discharge (ESD). The design of the Fuel Oxidation Tester shall preclude equipment damage due to ESD (internal/external electrical arcing), protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking in accordance with MIL-STD-1686C sections 5.2 and 5.7.2 and MIL-HBDK-263B for guidance.

2.18 Electro Magnetic Interference and Compatibility (EMI/EMC). The Fuel Oxidation Tester EMI and EMC emissions shall be in accordance with the following radiated emission and susceptibility requirements of MIL-PRF-28800F section 4.5.6.5. Emitted radiation absorption and radiation limits FCC Part 15.109 & 15.209.

2.19 Over Temperature Protection. The Fuel Oxidation Tester shall have an audible, visual or both methods of over temperature warning.

2.20 Operating Temperature Range. The Fuel Oxidation Tester shall meet the specifications listed herein with full capabilities within the temperature range of 68° F to 122°F (20°C to 50°C) after performing tests in MIL-PRF-28800F section 4.5.5.1.1 table 11 for class 3 equipment.

2.21 Storage Temperature Range. The Fuel Oxidation Tester shall meet and/or exceed manufacturer's specifications for stored equipment after tests of MIL-PRF-28800F section 4.5.5.1.1 table 11 for temperature range and relative humidity storage for class 3 equipment classification.

2.22 Altitude Requirements. The Fuel Oxidation Tester shall meet its operating performance and accuracy while operating at an altitude of 8,000 ft. (2,438 meters) ±2%. The equipment shall meet performance and accuracy requirements, for operation and non-operating storage, as specified in MIL-PRF-28800F sections 3.8.3 and 4.5.5.2 respectively at this altitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:31Z
Check the source to verify that this is the current version before use.

2.23 Calibration & Maintenance Adjustments. No special tools or software shall be required for alignment or adjustment except those furnished with the Fuel Oxidation Tester test set. A software license shall be issued for each physical instrument when the system is designed for computer based calibration. Any special equipment, fixtures, tools, standards, etc., required for calibration and not common to a secondary calibration laboratory shall be supplied with the Fuel Oxidation Tester. Any unique accessories including software required for calibration of this item shall be furnished by the manufacturer. Units that fail calibration and are recommended to be adjusted, either manually or using an automated alignment/adjustment software, the special hardware, manual instructions, software, software instructions and software licenses (where required by the manufacturer, and shall be valid for the life of the instrument) needed to perform the alignment/adjustment process shall be included with the Fuel Oxidation Tester. A list of peripheral support TMDE equipment required to accomplish the calibration/alignment/adjustments of the unit shall be provided with the Fuel Oxidation Tester. Any unique accessories including software required for calibration of this item shall be furnished by the manufacturer. All firmware or software provided under this purchase description shall be the same version, revision and revision date.

2.24 Calibration Period. The design of the Fuel Oxidation Tester shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the test set case or modules. The calibration interval shall be based on a period of one year or an operating time of 2,080 hours or whichever occurs first.

2.25 Reliability. The design of the equipment shall be such that under normal use and operation the mean time between failures (MTBF) exceeds the minimum of 2,080 hours of operation accordance with MIL-PRF-28800F section 3.13. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion in accordance with MIL-PRF-28800F section 3.13. The manufacturer shall be able to reasonably establish the claim that the offered equipment is capable of the required level of reliability by meeting and/or exceeding MIL-PRF-28800F sections 3.13 and section 4.5.8.1.

2.26 Maintainability. The Fuel Oxidation Tester shall meet the maintainability criterion in accordance with MIL-STD-1472G paragraphs 5.9 through 5.9.18 and 5.9.21 and MIL-PRF-28800F sections 3.14.1 through 3.14.4 and sections 6.6.2.4 and 6.6.24.1.

2.27 Preventive Maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800F sections 3.14.2 & 3.14.3.

2.28 Corrective Maintenance. The Fuel Oxidation Tester shall be designed to meet reparability criterion in accordance with section 3.14.3 sub sections a through d of MIL-PRF-28800F.

2.29 Electrical Power Sources & Connections. The Fuel Oxidation Tester shall operate from nominal commercial, military, aircraft and power sources with a nominal voltage of 120 VRMS Single Phase AC (±10%) at line frequencies of 50 Hz & 60 Hz (±5%), in accordance with MIL-PRF-28800F paragraph 3.1.5.2 and 3.1.5.2a. Transient state conditions shall meet MIL-PRF-28800F requirements for the classification specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:31Z
Check the source to verify that this is the current version before use.

2.30 Connectors. The system shall have a National Electrical Manufacturers Association (NEMA) type 5-15P AC power connector.

2.31 Weight. The system shall comply with MIL-STD-1472 section 5.9.11.3.3 for weight & size restrictions for single person lift & handling. The total weight of the Fuel Oxidation Tester, excluding accessories and manuals, shall not exceed a two-person mechanically assisted lift of 250 pounds, (114 kilograms).

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance. Minor changes to the commercial product, with respect to ruggedizing and environmental suitability and protection shall be permitted, subject to the approval of the procurement office, to satisfy the requirements of the procuring activity.

4.2 Alternative offers. A current, complete instrument with all items required for use, shall be offered/provided.

4.3 Serviceability, Reliability, and Quality of Materials.

4.3.1 All supplied items shall carry a standard commercial warranty on parts and labor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:31Z
Check the source to verify that this is the current version before use.

5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 ASTM documents may be obtained at at www.astm.org or from ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959.

6.1.2 FAR. FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/ .

6.1.3 Government Documents. Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://quicksearch.dla.mil or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg. 4, Section D, Philadelphia, PA 19111-5094. Phone (215) 697-1187, FAX (215) 697-2978, or Customer Service (215) 697-2667/2179.

6.1.4 NEMA documents. Application for copies should be addressed to NEMA, 1300 North 17th Street, Suite 1847, Rosslyn VA 22209, www.nema.org.

6.1.5 UL documents. UL Standards may be obtained at http://www.ul.com/ or from the Underwriters Laboratories, 333Pfingsten Road, Northbrook, IL 60062-2096.

6.2 National Stock Number NSN. The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with this CID.

NSN	6630-01-629-5112

6.3 Key Words.

Gas Turbine
High Temperature
Jet fuel

6.4 Changes from previous issue. Marginal notations are not used in this in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:31Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodian:
Air Force - 84

Preparing activity:
Air Force - 84

Reviewer:
Air Force - 99

Agent:
Air Force – 99

Project 6630-2015-001

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:31Z
Check the source to verify that this is the current version before use.